IUBAT- International University of Business Agriculture and Technology

Lab Report on

DisEase.AI – AI Based Disease Prediction System

Course Name: Software Engineering Lab

Course code: **CSC 470**

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Section: H

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Project Title:

DisEase.AI – AI Based Disease Prediction System

1. Project Description:

With the rapid growth of Artificial Intelligence (AI) and Machine Learning (ML), healthcare systems are becoming more intelligent and data-driven. Early disease detection and proper medical guidance are critical for saving lives and reducing healthcare costs. DisEase.AI is an AI-based disease prediction system designed to assist users by predicting possible diseases based on symptoms, medical reports, and personal information.

The system integrates Machine Learning (ML) for symptom-based disease prediction and Deep Learning for medical report analysis. It also provides doctor search functionality and downloadable medical summaries, making it a comprehensive digital healthcare assistant.

1.1 Objectives

The main objectives of this project are:

- To predict diseases based on user-provided symptoms using ML
- To analyze medical reports using Deep Learning
- To provide medical advice, treatment, and doctor suggestions
- To allow users to download prediction results as PDF reports
- To enable admin management of disease, doctor, and treatment data

1.2 System Benefits

- Early disease prediction and awareness
- Reduced dependency on immediate hospital visits
- Easy access to doctors' information
- Secure handling of medical data
- Time-efficient and user-friendly healthcare support

1.3 Software Process Model

The Incremental Software Development Model is used in this project. The system is developed in phases such as symptom prediction, doctor search, report analysis, and admin management.

1.4 Why Incremental Model is Chosen

- Allows gradual integration of ML and DL models
- Easy modification of requirements
- Early testing and validation of core features
- Suitable for AI-based academic projects

2. Project Requirements

2.1 User Requirements

The system has two primary actors: User (Patient) and Admin

User Requirements 01

Patients can input their symptoms, personal details, and duration of illness to receive AI-based disease predictions with treatment recommendations.

System Requirements:

1. The system must validate that at least 3-5 symptoms are selected
2. The system must process inputs through trained ML models
3. The system must display prediction results with confidence scores
4. The system must generate downloadable PDF reports

Functional Requirements:

1. User navigates to prediction page
2. User enters personal details (Name, Age, Sex, Days Suffering)
3. User selects symptoms from predefined list or enters custom symptoms
4. System processes data through ML prediction engine
5. System displays predicted disease with treatment plan
6. User can download complete report as PDF
7. System stores prediction history for future reference

User Requirement 02

Users can search for doctors based on multiple parameters and view detailed contact information.

System Requirements

1. The system must support multi-parameter search functionality
2. The system must retrieve doctor data from secure database
3. The system must display search results in user-friendly format
4. The system must protect doctor's contact information from misuse

Functional Requirements:

1. User accesses doctor search page
2. User applies filters (Location, Hospital, Specialty, Disease)
3. System queries database and displays matching doctors
4. User views doctor details (Name, Contact, Chamber, Hospital)
5. System provides direct contact options (call/email)

User Requirement 03

Users can upload medical reports for AI-based analysis and receive

System Requirements:

1. The system must support multiple medical image formats
2. The system must process images through CNN models
3. The system must generates comprehensive analysis reports
4. The system must ensure data privacy and security

Functional Requirements:

1. Users upload medical reports (ECG, X-ray, Blood Test, etc.)
2. System validates file format and size
3. CNN model analyzes the report
4. System displays abnormalities and recommendations
5. User can download analysis report
6. System stores reports for future reference

2.2 Admin Requirements

User Requirement 01

Admin can manage all system data including diseases, doctors, and user activities.

System Requirements:

1. The system must provide secure admin authentication
2. The system must supports CRUD operations on all data entities

3. The system must maintain activity logs
4. The system must ensure data integrity and consistency

Functional Requirements:

1. Admin logs in with secure credentials
2. Admin can add/edit/delete disease entries
3. Admin can manage doctor database
4. Admin can upload bulk data via CSV
5. Admin can view user activity logs
6. Admin can monitor system performance

User Requirement 02

Admin can update and retrain AI models based on new medical data

System Requirements:

1. The system must support model versioning
2. The system must validate new model performance
3. The system must ensure seamless model deployment
4. The system must maintain prediction accuracy logs

Functional Requirements:

1. Admin uploads new training datasets
2. System retrains ML/DL models
3. Admin validates model performance
4. System deploys updated models
5. Admin monitors prediction accuracy trends

Non-Functional Requirements

- Prediction response time < 10 seconds
- Secure data storage using Django ORM
- Scalable system design
- Consistent ML/DL model output
- Responsive UI for desktop & mobile
- Modular Django architecture

Function Point Analysis:

Functionality Input Output:

No	Functionality	Input	Output
1	User Enter Disease Symptoms	Symptoms (3–5), Name, Age, Sex, Suffering	Input Validation Status
2	Predict Disease	Symptoms, User Details	Predicted Disease Name
3	Generate Treatment & Advice	Predicted Disease	Treatment Plan, Primary Medicines
4	Generate Doctor Suggestion	Disease Name, Location (optional)	Doctor Recommendation
5	View Prediction Result	Prediction Request	Disease, Treatment, Advice Display
6	Download Prediction Report	Prediction Result Data	PDF File (Disease & Advice)
7	Search Doctor Information	Doctor Name / Disease / Hospital / Location	Doctor List
8	View Doctor Details	Doctor ID / Selection	Doctor Profile Information
9	Upload Medical Report	ECG / X-ray / Blood / Urine Report	Upload Status
10	Analyze Medical Report (CNN)	Uploaded Report File	Medical Report Summary
11	Generate AI Interpretation	CNN Output	Diagnosis Hints & Abnormalities
12	Download Medical Report Summary	Analysis Result	PDF Report with Date & Summary
13	Admin Login	Admin Email, Password	Admin Dashboard Access
14	Manage Doctor Database	Doctor Info (Add/Edit/Delete)	Update Confirmation
15	Manage Disease Database	Disease & Medicine Data	Data Update Confirmation
16	Bulk Upload Doctor List	CSV File	Upload Success / Error
17	View User Activity Logs	Date Range / Filter	Activity Log Report
18	Monitor Model Performance	Model Logs	Performance Statistics
19	Generate System Reports	Report Type, Time Period	PDF / Excel Report
20	Backup System Data	Backup Request	Backup Confirmation

Fn.In.Out | Identify Complexity (TF) | Identify Complexity (DF) | UFP (TF) | UFP (DF) | TDI | FIN

Identify Complexity (TF):

No	Transaction Functions	TF	Fields / File Involvement	FTR	DET
1	User Enter Symptoms & Personal Info	EI	Fields – Name, Age, Sex, Symptoms, Suffering Days; File – User_Ir	1	5
2	Disease Prediction	EO	Fields – Symptoms; Files – Disease_DB, ML_Model	2	4
3	Generate Treatment & Advice	EO	Fields – Disease Name; Files – Treatment_DB, Medicine_DB	2	4
4	Generate Doctor Suggestion	EO	Fields – Disease Name, Location; File – Doctor_DB	1	4
5	View Prediction Result	EQ	Fields – Disease, Treatment, Advice; File – Prediction_Result	1	4
6	Download Prediction PDF	EO	Fields – User Info, Disease, Advice; File – PDF_Report	1	4
7	Search Doctor Information	EQ	Fields – Doctor Name, Hospital, Location, Disease; File – Doctor_D	1	4
8	View Doctor Details	EQ	Fields – Doctor ID, Name, Contact, Hospital; File – Doctor_DB	1	4
9	Upload Medical Report	EI	Fields – Report File (ECG/X-ray/Blood); File – Medical_Report	1	3
10	Analyze Medical Report (CNN)	EO	Fields – Uploaded Report; Files – CNN_Model, Medical_Report	2	4
11	Generate Report Summary	EO	Fields – Analysis Output; File – Report_Summary	1	3
12	Download Medical Report PDF	EO	Fields – Summary, Date; File – PDF_Report	1	3
13	Admin Login	EI	Fields – Admin Email, Password; File – Admin	1	2
14	Admin Manage Doctor Data	EI	Fields – Doctor Info; File – Doctor_DB	1	4
15	Admin Manage Disease Database	EI	Fields – Disease, Medicine Info; File – Disease_DB	1	4
16	Bulk Upload Doctor List	EI	Fields – CSV File; File – Doctor_DB	1	2
17	View User Activity Logs	EQ	Fields – Date Range, Activity Type; File – Log_DB	1	3
18	Monitor Model Performance	EQ	Fields – Model Logs, Accuracy; File – Model_Log	1	3
19	Generate System Report	EQ	Fields – Report Type, Date Range; Files – User, Disease, Report	3	3
20	Backup System Data	EO	Fields – Backup Request; Files – All_DB	3	2

Fn.In.Out | Identify Complexity (TF) | Identify Complexity (DF) | UFP (TF) | UFP (DF) | TDI | FINAL CALC

Identify Complexity (DF):

No	Data Functions	Fields / File Involvement	RET	DET
1	User	User_ID, Name, Age, Sex, Email, Role	1	6
2	Patient Input Data	Input_ID, User_ID, Age, Sex, Symptoms, Duration_D	1	6
3	Symptom Database	Symptom_ID, Symptom_Name, Description	1	3
4	Disease Knowledge Base	Disease_ID, Disease_Name, Description, Risk_Level	1	4
5	Prediction Result	Result_ID, Input_ID, Disease_Name, Probability, Pre	1	5
6	Treatment Information	Treatment_ID, Disease_ID, Treatment_Details	1	3
7	Medicine Information	Medicine_ID, Disease_ID, Medicine_Name, Dosage	1	4
8	Doctor Suggestion	Suggestion_ID, Disease_ID, Advice, Precaution	1	4
9	Prediction Report	Report_ID, User_ID, Disease_Name, Date, Download	1	5

Unadjusted Function Point (TF):

No	Transaction	Type	FTR	DET	Complexit	UFP	
1	User Enter	EI	1	5	Low	3	
2	Disease Prediction	EO	2	4	Low	4	
3	Generate Treatment	EO	2	4	Low	4	
4	Generate Doctor	EO	1	4	Low	3	
5	View Prediction	EQ	1	4	Low	3	
6	Download	EO	1	4	Low	3	
7	Search Doctor	EQ	1	4	Low	3	
8	View Doctor	EQ	1	4	Low	3	
9	Upload Medical	EI	1	3	Low	3	
10	Analyze Medical	EO	2	4	Average	4	
11	Generate Report	EO	1	3	Low	3	
12	Download Medical	EO	1	3	Low	3	
13	Admin Login	EI	1	2	Low	3	
14	Admin Manage	EI	1	4	Low	3	
15	Admin Manage	EI	1	4	Low	3	
16	Bulk Upload	EI	1	2	Low	3	
17	View User Activity	EQ	1	3	Low	3	
18	Monitor Model	EQ	1	3	Low	3	
19	Generate System	EO	3	3	Average	5	
20	Backup System	EO	3	2	Low	4	
							61

Fn.In.Out | Identify Complexity (TF) | Identify Complexity (DF) | UFP (TF)

EI Table

FTR's	DATA ELEMENTS		
	1-4	5-15	> 15
0-1	Low	Low	Ave
2	Low	Ave	High
3 or more	Ave	High	High

Shared EO and EQ Table

FTR's	DATA ELEMENTS		
	1-5	6-19	> 19
0-1	Low	Low	Ave
2-3	Low	Ave	High
> 3	Ave	High	High

Values for transactions

Rating	VALUES		
	EO	EQ	EI
Low	4	3	3
Average	5	4	4
High	7	6	6

1

Unadjusted Function Point (DF):

No	Data Functions	Fields / File Involvement	RET	DET	Complexi	UFP	Total UFP
1	User	User_ID, Name, Age, Sex,	1	6	Low	7	
2	Patient Input Data	Input_ID, User_ID, Age, Sex,	1	6	Low	7	
3	Symptom Database	Symptom_ID,	1	3	Low	7	
4	Disease Knowledge	Disease_ID, Disease_Name,	1	4	Low	7	
5	Prediction Result	Result_ID, Input_ID,	1	5	Low	7	
6	Treatment	Treatment_ID, Disease_ID,	1	3	Low	7	
7	Medicine Information	Medicine_ID, Disease_ID,	1	4	Low	7	
8	Doctor Suggestion	Suggestion_ID, Disease_ID,	1	4	Low	7	
9	Prediction Report	Report_ID, User_ID,	1	5	Low	7	
							63

RET's	DATA ELEMENTS		
	1-19	20 - 50	> 50
1	Low	Low	Ave
2-5	Low	Ave	High
> 5	Ave	High	High

Rating	Values	
	ILF	EIF
Low	7	5
Average	10	7
High	15	10